

Kavit Jay Mandalaywala

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Education

The Pennsylvania State University

College of Engineering — Bachelor of Science in Computer Science

College of Science — Bachelor of Science in Mathematics

University Park, PA

Expected 2028

GPA: 3.72/4.00

Professional Experience

RWadiwala Technologies

Surat, Gujarat

Trading Intern

Jun 2026 – Present

- Conducted market research and analyzed equity price movements using quantitative and technical indicators.
- Evaluated trading opportunities through data-driven analysis, risk assessment, and portfolio monitoring.
- Tracked market trends and performance metrics to support investment decision-making.

Penn State Learning

University Park, PA

Math Tutor

Aug 2025 – Present

- Tutored students in Calculus, Linear Algebra, and introductory Probability through one-on-one and group sessions.
- Assisted students with homework, exam preparation, and problem-solving strategies.
- Explained complex quantitative concepts clearly, strengthening communication and teaching skills.

Nittany Market Analysis Association (NMAA)

University Park, PA

Club Member

Aug 2024 – May 2025

- Conducted equity and macroeconomic research through team-based stock pitch projects.
- Applied quantitative analysis to evaluate company fundamentals, market trends, and investment opportunities.
- Presented investment theses to peers and mentors, strengthening financial communication skills.

Other Experience

Nittany Lion Consulting Group (NLCG)

University Park, PA

Consulting Training Program Participant

Jan 2025 – Dec 2025

- Completed consulting training focused on structured problem solving, client engagement, and presentation.
- Developed analytical reasoning and teamwork skills through case studies and group presentations.

Nittany Data Labs (NDL)

University Park, PA

Club Participant

Aug 2024 – Dec 2024

- Learned Python libraries including NumPy, Pandas, and Matplotlib for data analysis and visualization.
- Strengthened data science foundations for quantitative finance, research, and analytics projects.

Projects

Microstructure Alpha Engine

Jun 2026 – Present

- Built a real-time microstructure platform streaming live order book data via Alpaca WebSocket, implementing Kyle's Lambda, Almgren-Chriss optimal execution, and Order Flow Imbalance signal generation.
- Estimated price impact via rolling OLS on signed order flow; solved optimal execution trajectory using quadratic optimization, minimizing market impact cost vs. VWAP benchmark.
- Decomposed bid-ask spreads into information, inventory, and processing components using Glosten-Milgrom approximation; visualized models live in a Bloomberg-style terminal dashboard.

Market Regime Detection & Trading Strategy

May 2026 – May 2026

- Built a Hidden Markov Model to classify bull and bear market regimes using historical equity return data.
- Developed regime-based trading strategies and evaluated performance using Sharpe ratio, volatility, and drawdowns.
- Analyzed state transitions and market behavior to improve portfolio allocation and risk management decisions.

Algorithmic Trading Dashboard

Mar 2026 – Apr 2026

- Developed a multi-page Streamlit application connected to Alpaca's paper trading environment.
- Structured the codebase across backtesting, signals, risk, and execution modules for scalability.
- Implemented portfolio analytics including equity curves, drawdowns, turnover, and performance metrics.

Intraday Microstructure Alpha & Execution Simulator

Sep 2025 – Sep 2025

- Built a synthetic limit order book simulator to model market microstructure dynamics.
- Engineered order book imbalance, microprice drift, and trade-flow signals for alpha generation.
- Simulated intraday trading behavior to study execution quality, liquidity, and short-term price movement.

Skills

Programming: Python, C, MATLAB, Git, Linux/Unix, Streamlit, Alpaca API, yFinance

Data Science: NumPy, Pandas, Matplotlib, Data Cleaning, Data Visualization, EDA, API Integration

Quant Finance: Portfolio Optimization, Modern Portfolio Theory, Monte Carlo Simulation, Risk Modeling, Sharpe Ratio, Volatility, Drawdowns, Factor Signals

Trading and ML: Market Microstructure, Order Book Simulation, Supervised Learning, Time-Series Forecasting, RSI, MACD, Bollinger Bands

Mathematics: Calculus, Linear Algebra, Probability, Statistics, Optimization, Equity Research, Macroeconomic Research